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RESEARCH

A Congestion-Aware Routing System in Wireless Sensor Networks Based on Bee Colonies and Intelligent Butterfly Optimisation

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Abstract
Data is sent by multi-hop communication in Wireless Sensor Networks (WSNs), where each sensor node transmits data to its neighbouring nodes until it reaches the target node, which could be a base station or sink node. Congestion is a potential when network size and data traffic both rise. When the network is overloaded with data, it is said to be in congestion, which results in packet collisions, higher latency, and worse network performance overall. Numerous congestion-aware routing techniques have already been created in WSNs to address this limitation. By intelligently routing data traffic through less congested pathways and avoiding densely populated areas, congestion-aware routing systems seek to mitigate congestion-related concerns. However, in order for the network's congestion-aware routing algorithms to collect and distribute congestion information, they need extra overhead or control messages. This additional overhead, particularly in sensor nodes with limited resources, might result in higher energy consumption and worse network performance. This work established a new congestion-aware routing protocol called ButBeeRoute in order to address this key shortcoming. However, ButBeeRoute is designed with dynamically arranged node clusters to offer complete coverage and connectivity. This research also employs novel cluster-based WSNs to reduce packet loss and save energy. When the network is made up of nodes with varying transmission ranges and the least amount of power consumption, however, ButBeeRoute can determine which method is optimal or most dependable. The simulation results demonstrate that compared to other traditional protocols, ButBeeRoute has a reduced packet drop rate, increasing the ratio of packet distribution, network life, and residual energy. Comparing the ButBeeRoute strategy to the current one, it resulted in an overall improvement of 13% in packet loss rate, 20% on average energy consumption, and 18% on average reduced traffic.

Keywords Artificial bee colony · Cluster head · Wireless sensor networks · Congestion control · Congestion occurrence rate · Bee optimization · Butterfly optimization

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1 Introduction

A wireless sensor network (WSN) comprises a group of shared sensor nodes that are all located within or near the occurrence. The number of nodes might range from thousands to millions, depending on the application. A sensor, a processor, a transmitter, and a power supply unit are the four essential components of the sensor unit [1, 2]. The sensor unit ought to be low-cost, battery-operated, effortless to set up, self-configuring, and put in a location where it will be exposed to severe conditions [3, 4]. Thermal and infrared radar are two types of sensors that can detect ambient factors such as temperature, pressure, sound, and humidity [5]. Sensors, on the other hand, are hampered by a lack of memory, power, computing resources, and dependability [5, 6]. In WSN, congestion is a big issue [7–9]. It occurs when the load exceeds the capacity of the node, when the wireless channel is shared by numerous nodes, resulting in a collision, or when the channel is reduced, resulting in a loss in bandwidth. Congestion causes an increase in the percentage of lost revenue, a worsening of the unfair traffic channel's quality, increased delay, and power loss. It also completes the relay expansion while lowering bandwidth and network life. As a result, congestion in the WSN must be reduced.

Congestion reduction is usually divided into two stages: detection and management. When packets are transported from one node to another, events can occur [10, 11]. Congestion detection is the process of identifying and recognizing areas or nodes in the network that are experiencing high levels of data traffic or packet queuing, leading to potential performance degradation [12, 13]. On the other hand, congestion management refers to the strategies and mechanisms employed to alleviate or mitigate congestion once it is detected [14]. Congestion-aware routing protocols are one of the approaches to congestion management.

Routing protocols must limit the number of packets on the network to reduce congestion. Dropping leaky packets, on the other hand, reduces data accuracy and increases power consumption. Conventional congestion management methods often employ a back-pressure method to alleviate congestion by lowering the number of child node transfers at congestion nodes and the packet creation rate at sensor nodes, both of which result in increased traffic [15]. Existing congestion-aware routing protocols require additional control messages or overhead to gather and disseminate congestion information across the network. This extra overhead can lead to increased energy consumption and reduced network efficiency, especially in resource-constrained sensor nodes. Many existing congestion-aware routing protocols may not scale well as the network size increases. As the number of nodes grows, the overhead and computational complexity of congestion estimation and management may become prohibitive, affecting the overall scalability of the protocol. An ideal method of pathfinding is required to solve these issues. The ButBeeRoute technique was thus introduced in this study. The main contributions of this paper are as follows:

(i) To resolve and reduce congestion in the WSN using the ButBeeRoute optimization method. ButBeeRoute involves distributing data traffic evenly across the network to avoid overloading specific nodes or cluster regions.

(ii) Analyzing and improving the quality of WNS by reducing packet loss through ButBeeRoute. ButBeeRoute Predicting traffic patterns and congestion-prone areas can help in proactively avoiding those regions and diverting traffic elsewhere.
(iii) Reduce the execution time of the optimization method by incorporating the butterfly and bee approaches. Implementing congestion-aware routing protocols that can dynamically adjust their paths based on real-time congestion feedback.
(iv) To improve the lifetime of WSN by conserving energy efficiency and limiting the rate of data transmission to prevent network overload. Efficiently managing data queues at each node to reduce packet loss and improve overall performance by using ButBeeRoute.

Application of the congestion control strategies outlined in the WSN, along with analysis and comparison of the outcomes to existing approaches to show the effectiveness of the proposed approach, is another key addition.

2 Related Works

The literature review of numerous research publications related to congestion avoidance and congestion management algorithms is explained in this part.

In [14], a unique routing approach is presented that reduces congestion. There are three primary modules in the adaptive routing technique. First, using multistage fuzzy logic, an ideal notification level for queue control is determined. Congestion Control management with adaptive buffer switching has been invented [15]. The aforementioned approach demonstrates how to select the primary and backup buffers in an adaptive manner based on the cost assessment. Congestion is improved by dynamic buffer switching and swapping.

Adaptive cuckoo search (ACS) [16], artificial bee colony behavior, wolves and herds of fish, and algorithms for gravitational search and optimization of ant colonies are examples of naturally inspired optimization techniques. According to the load on the system, Pan et al.'s adaptive BBR congestion window [17] changes the congestion window gain of each BBR flow. The suggested algorithms deal with the RTT fairness issue by controlling the upper bound of the congestion window in accordance with the delivery rate and queue state. The design of a multipath protocol is suggested in [18] using the particle swarm optimization technique. It is intended to improve QoS metrics and WSN efficiency.

An energy-conscious QoS MAC system based on classified data and multi-hop forwarding has been presented by Sakib et al. in [19]. Kafi et al. proposed REFIACC. It is a congestion-management technique that prevents interference in interstitial and in-road hotspots by using reliable, efficient, and interference-free communication planning. The maximum bandwidth consumption is used, which surpasses the total performance [20].

Makarem et al. [21] improved the retransmission timeout computation for congestion identification by implementing two congestion controls. It successfully combats congestion while maintaining its relatively constrained device demand by using a rate-based method. An efficient congestion reduction solution is developed in [22] using ant colony optimization and the Huffman coding technique to increase network efficiency. It combines traffic- and resource-oriented optimization to find a ton of alternate, congested-free paths.

Amit Grover et al. [23] propose a novel rate-aware congestion management strategy that distinguishes between three different types of congestion. Bibin Christopher and Jasper [24] introduce a Jellyfish Dynamic Routing Protocol for maintaining location privacy and congestion avoidance with guaranteed lower latency. This routing algorithm subdivides the entire sensor field into several subdivisions, and each group selects a target site by calculating its transmission distance. The rate-based congestion management strategy is presented in [25] in order to reduce energy usage. This method relies on cluster routing to function. A rate control system reduces the end-to-end delay over a long simulation time to extend network life. To reduce power consumption and packet loss, El Alami and Nagid are working on routing techniques that take receiver mobility into account as a significant issue. A method known as routing-Gi has turned out to be more effective than earlier options in terms of network life, energy efficiency, and packet distribution speed [26].

To extend network life, El Alami and Nagid are using a new method to cluster topologies, with high-power nodes collecting data to send to the BS. This strategy also addresses the concern of gathering node information that is no longer relevant. It saves energy by operating on a sleep–wake cycle. In the same network as heredity [27], this method is utilized. In WSN, Sangeetha et al. developed heuristic ways for locating the most energy-efficient paths. To save battery power, the writers focus on regulating the level and top of the node on a regular basis. To avoid congestion, the information flow is then unbiased through ambiguous logic. If there is a bottleneck, he uses real-time learning and stellar memory techniques to find the best solution [28].

Logambigai et al. Work on heuristic techniques has continued, with a new energy-efficient network-based cluster approach with intelligent fuzzy rules being introduced. Routing is done with the help of a network coordinator who has unclear rules and takes into account the smallest number of middle nodes in the relay procedure. This strategy is efficient in terms of grid life as well as energy [29]. Sangeetha et al. proposed another enhanced congestion-aware routing technique based on an ambiguous set of rules. This is a logical traffic management strategy that discovers more dependable routes and handles extra traffic using hazy legal projections. It has two parts: the first is to set a path by locating non-local nodes, and the second is to soften the way without causing congestion. The ECFM algorithm [30] was used to estimate traffic congestion.

Selvi et al. developed another clustered gravitational and confusing energy-based strategy. Existing transportation systems' limitations and issues must be addressed. It employs a clockwise gravity clustering algorithm to give optimal routing and clustering solutions. After applying a hazy set of principles, the path that best suits the cluster head node is chosen. ACO [31] is a naturally occurring metaheuristic mechanism that exhibits many of the fine-scale features. Meta-heuristic strategies are more successful at minimizing congestion and are crucial for large-scale networks. The ACO's mission [32–36] is to discover alternatives to traffic congestion. In addition, the heuristic values of the various paths play an essential part in determining the best path. When incomplete information is presented in a timed format, Janes extends Shannon's approach by calling it the original principle of maximizing optimal entropy measures [37–40].

Estimating congestion accurately is a challenging task, especially in dynamic and unpredictable WSN environments. Inaccurate congestion estimation can lead to suboptimal

routing decisions, exacerbating congestion issues rather than alleviating them. The above-mentioned congestion detection and management approaches are affected by dynamic and unpredictable WSN environments. So it leads to insufficient congestion accuracy. This problem is solved by the proposed ButBeeRoute method.

3 Proposed Work

The purpose of this article is to avoid congestion situations and to look for congestion in its early stages to eliminate the worst consequences of congestion in the network. The effects of congestion are packets that slow down the increase in road traffic, slow down processes, and lower network bandwidth. Limited resources such as node power, bandwidth, and storage are also causes of WSN congestion.

ButBeeRoute is a load-balancing and congestion-aware routing algorithm. It is based on two main concepts. One option is to divide the network nodes into three priority groups based on their network's center position and then choose alternative routing algorithms for nodes in different regions. Another option is to select the output node based on distant overload information.

In this paper, we'll concentrate on managing network congestion, think about minimizing power usage and network slowdowns, and create active road congestion algorithms to handle congestion. We assume the logical assumptions below to streamline the WSN model:

Table 1 Shows list of abbreviation and its description which are used in this work

Abbreviation	Description
WSNs	Wireless sensor networks
QoS	Quality of service
ACS	Adaptive cuckoo search
PSO	Particle swarm optimization
BA	Base station
CH	Cluster head
SN	Sensor nodes
BOA	Butterfly optimization algorithm
RE	Residual energy
NT	Node's traffic
DBS	Distance to BS
DSN	Distance to the sensor nodes
NC	Node centrality
ND	Node degree
FF	Final fitness
QL	Queue length
SC	Selection score
ABC	Artificial bee colony
COR	Congestion occurrence rate

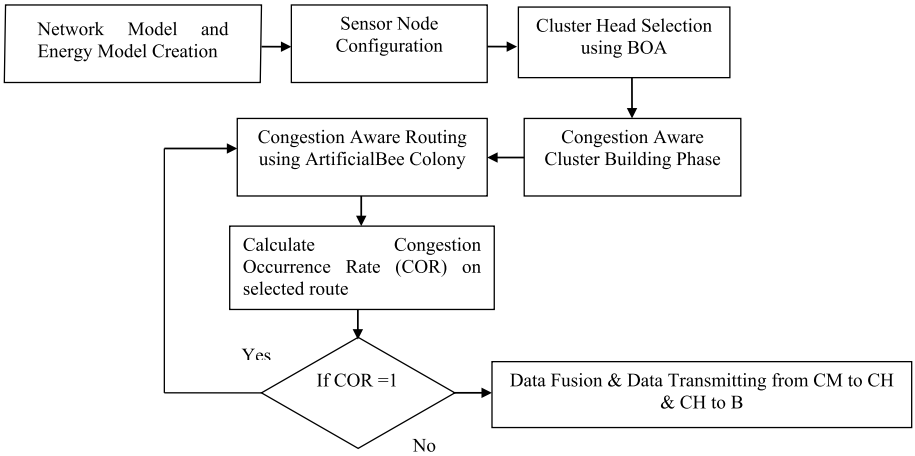


Fig. 1 Network model of proposed work

After deployment, an array of sensor nodes is quietly scattered at random in a particular region. All sensor nodes have the same routing and information-gathering abilities and can calculate their distance from the source node according to the quality of the received signal (Table 1).

3.1 Network Model

The network model assists in avoiding congestion and maximizing WSN power efficiency. Figure 1 depicts the network model of the proposed work, illustrating the structure and components involved in the ButBeeRoute algorithm. It visually represents the arrangement of nodes within the network, showcasing how the algorithm operates to mitigate congestion and optimize data routing. The following factors are taken into account when creating a network template:

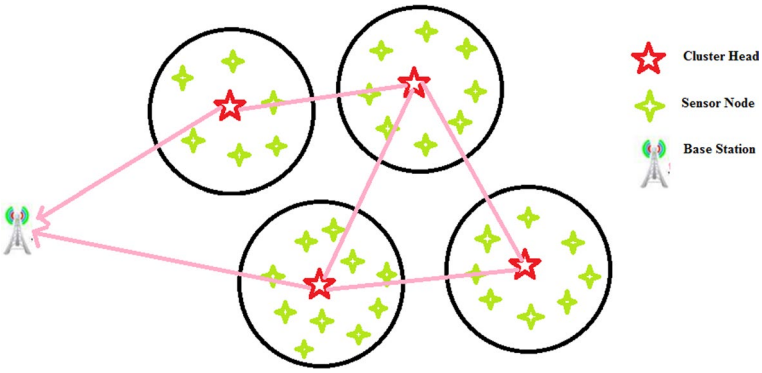


Fig. 2 Network model of proposed work

- The initial power and processing time of all sensor nodes in a WSN are similar, and the distance between them is computed using the Euclidean distance formula.
- The sensor unit is placed in the sensor region at random, and the sensor's position is set after deployment.
- BS receives residual power and information from the sensor unit through a remote connection.

An effective channel selection technique is used to pick channels for all sensor nodes based on this information. After that, the routing mechanism is employed to determine the path between channels and the BS (Fig. 2).

3.2 Energy Model

The energy expended in transmitting a qth bit to a distance 'd' and getting a qth bit from a distance of 'd' is denoted by E_s and E_r energy dissipation is E_{ed} as well as energy amplification is E_{ae}.

$$E_s(q,d) = q * E_{ed} + q * E_{ae} d^2 \tag{1}$$

$$E_r(q) = q * E_{ed} \tag{2}$$

3.3 Cluster Head Election Using BOA

The butterfly identifies the sensor group to pick as CH from the network sensors during the BOA CH election phase. The quantity of CHs in the network is identical to every dimension of the butterfly. A random node identification between nodes and the overall number of nodes within the network determines the position of each butterfly. In the CH selection procedure, the improvisation of the butterfly position is taken into account.

By computing the appropriate value, the BOA chooses the best channel among all the sensors. The butterfly odor is calculated using the fitness function. The strength of the butterfly's stimulation is determined by the fitness function, and the fragrance of the butterfly is estimated to keep it in position. To compute acceptable values, the traffic node level, the middle node of the distance node to the neighbor, the distance to the BS, and the remaining power are employed.

1. Residual energy (RE): The entire amount of transferred joules per joule is the energy efficiency. Calculate a node's average power. The head of the cluster is more conditionally selected than a node with a power level higher than or equivalent to the average power cost of the cluster. Free energy is considered negligible. The residual energy (RE) is described in Eq. (3).

$$\text{RE} = \frac{\text{Total energy of cluster}}{\text{Total noof nodes}} \tag{3}$$

2. Node's traffic (NT): Used for node transfer, try to assess the node's queue capacity range. If node traffic is less than or equivalent to the common load rate, cluster heads are more likely to be chosen. The following equation depicts node traffic (NT) (4).

$$\text{NT} = \frac{\text{Total noof nodes in cluster}}{\text{Total noof nodes}} \tag{4}$$

3. Distance to BS (DBS): The distance between CH and BS is determined. The power utilization of a node is related to the length of the transmission path. It will take more electricity to send data if the BS is placed far away from the CH. As a result of the increased power consumption, an abrupt drop in CH may occur. As a result, during data transmission, nodes that are closer to the BS are chosen. The distance to BS (DBS) is represented by the equation: (5) which is crucial for understanding the power utilization of nodes based on the length of the transmission path. It highlights the significance of selecting nodes closer to the BS to minimize power consumption during data transmission and mitigate the risk of abrupt power depletion in CHs.

$$\text{DBS} = \frac{\text{Total distance in cluster}}{\text{Total noof nodes}} \tag{5}$$

4. Distance to the Sensor Nodes (DSN): The distance between a common sensor node and its CH is calculated. When the selected node has a shorter transmission distance to the BS, the node's power consumption is minimal. The Eq. (6) shows the distance to the sensor node (DSN) which emphasizes the importance of considering the transmission distance to the BS when selecting nodes. Nodes with shorter transmission distances to the BS consume less power, making them preferable choices for data transmission.

$$\text{DSN} = \frac{\text{Total distance in each cluster}}{\text{Total noof nodes}} \tag{6}$$

5. Node Centrality (NC): Node centrality (NC) determines the number of nodes centered by adjacent nodes and is shown in Eq. (7).

$$\text{NC} = \frac{\text{Total centerality in cluster}}{\text{Total noof nodes}} \tag{7}$$

6. Node Degree (ND): It establishes the quantity of sensor nodes associated with each CH. Because CNs with a higher number of cluster members lose their power in a shorter amount of time, CNs with a smaller number of sensors are chosen. The equation shows the node degree (ND) (8).

$$\text{ND} = \frac{\text{Total number of nodes in cluster}}{\text{Total noof nodes}} \tag{8}$$

Finally, using the Eq. (9), each node's fitness value BOS is computed as

$$\text{FF} = \text{RE} + \text{NT} + \text{DSB} + \text{DSN} + \text{NC} + \text{ND} \tag{9}$$

The procedure for selecting CH using BOA is described in detail above, and its pseudocode is supplied in "Algorithm 2."

Cluster head selection using BOA

```
Input:  All Sensor Nodes N
Output: Cluster Head Node CH
Variables:
Residual energy -RE
Node's traffic-NT
Distance to BS-DBS
Distance to the Sensor Nodes-DSN
Node Centrality-NC
Node Degree-ND
Produce a preliminary population of butterflies.
    for p= highest amount of iterations
        for all butterfly in population
            Calculate fragrance through eq. 9.
        end
        find the best butterfly population
        for all butterfly in population
            Produce a random value from [0,1]
                If  population found
                    Move on the way to an optimal node
                else
                    Move arbitrarily
                end
        end
        Update the value optimal node
    end
```

3.4 Congestion Aware Clusters Building Phase

This stage generates several clusters based on the CH selected. The update table contains information about cluster bottlenecks caused by surrounding nodes. Each node calculates its own weight and sends it to its neighbors, who use it to update their update tables. The update table has the following fields:

(ID of the node, present queue length, remaining energy, level of congestion).

Where the node ID is the neighbor's ID, the present queue size is the node's current range length, and each time a packet is queued or closed, it increases or decreases by one. The node's leftover energy is called residual energy, and the congestion level is determined by comparing the existing queue length and residual energy to the congestion occurrence score value.

After selecting each CH, the CH broadcasts an advertising package to the neighboring node. Non-CH nodes choose CH advertising packages with a lower Selection Score (SC) value to join a specific CH. The SC values are calculated based on Eqs. (10) and (11).

$$SC = |dis(CH, nonCH)| + CE + QL \tag{10}$$

where CE represents the CH node's spent energy, QL represents the CH node's current queue length, and dis(CH, nonCH) represents the distance between CH and nonCH.

When an adjacent node comes to the decision to connect a particular CH, it sends a one-time message with its own detect and the find of the CH in question. CH is responsible for allocating TDMA time to the cluster's members and broadcasting it on the broadcast schedule after the cluster has been created. This permits the node to only wake up and send data during the TDMA time slot. The bee colony routing protocol is combined with the TACMA-based MAC protocol (Medium Access Control).

4 Congestion Aware Routing using Artificial Bee Colony

The meta-heuristic method Artificial Bee Colony (ABC) takes cues from honey bee behavior and outperforms other algorithms in terms of functional optimization. ABC is used to solve isolated problems, which are represented as graphs with nodes and connections.

1. Each node in this setup had a certain number of bees. To make a trail, a bee is placed in each CH. Worker bee packs are packets created by sources.
2. Based on the likelihood matrix, employee bee packets are transferred to the subsequent CH node at random. This transmission will continue until the destination node receives the employee bee packet.
3. The worker bee chooses the next hop based on the path selection parameter, which reflects the likelihood of the node being chosen as the bee's next destination.

Congestion Avoid Algorithm

```
Input:  incoming packet
Output: incoming packet is inserted in queue / packet is dropped
Variables:
LowT-Lower threshold
HighT - Upper threshold
QL- Current queue length
For all node
        For each packet to queue in confirm QL against LowT and HighT
                If (QL< LowT)
                        Incoming packet is stored queue with low priority
                Else if (QL> LowT) && (QL< HighT)
                        Incoming packet is stored queue with high priority
                Else
                        Drop incoming packet
        Endif
End
```

4.1 Route Selection Parameters

The ABC method presented in this paper is the path distribution protocol. Each node selects the nearest node to receive a packet based on the path select variable after calculating the path variable depending on the information at hand. The ABC method defines

Table 2 Congestion occurrence rate (COR) Table

Condition	COR	Action
$QL <$ LowT with high RE	1	ready to forward
$QL <$ LowT with medium RE	2	ready to forward
$QL <$ LowT with low RE	3	forward packets based on priority
$QL >$ LowT && $QL <$ HighT with high RE	4	forward packets based on priority
$QL >$ LowT && $QL <$ HighT with medium RE	5	forward packets based on priority
$QL >$ LowT && $QL <$ HighT with low RE	6	*R* forward packets based on priority
$QL >$ HighT with high RE	7	drop packets
$QL >$ HighT with medium RE	8	drop packets
$QL >$ HighT with low RE	9	drop packets

four path selection criteria to evaluate the nodes on all three scales for distance, processing capacity, and congestion. The relationship between a node's separation from a receiver and the distance from a different node is known as the "forward rate". "Network load" represents the ability of a node to process data packets. "The cache remaining rate" and "the forward average cache remaining rate" are used to represent the node's perception of the level of congestion in and around it.

Forward Rate: Due to sensor nodes' limited transmission range, many nodes tend to be unable to interact with the receiving node directly. It forces the deployment of additional nodes to send data. As a result, the protocol for routing must take into account the spatial distance between the front node, the nearby node, and the user's node when choosing a route.

Network Load: The active node load factor of a node describes its performance and its capacity to receive packets. For nodes with a high load factor, the data transmission capacity is good. The node may send and get more packets since its packet cache range has a shorter lifetime. The routing algorithm should refrain from sending excess messages to low-load nodes in order to prevent network congestion and packet loss.

Cache Remaining Rate: Event-driven WSN data transfer techniques reveal a variety of aspects that could clog the network, increase latency, and hinder the efficiency of transmissions during a traffic explosion. In order to avoid excessive cache nodes, the method of routing must take into consideration the possibility that filling the cache queue may be a reflection of the node's storage condition. Additionally, it must shift the traffic load to a node that is idle or possesses a low caching level in order to use the network's wireless resources equitably.

Forward Average Cache Remaining Rate: The residual cache rate of the nodes themselves may give a precise indicator of how crowded it is. However, focusing only on the nodes is insufficient. Once a node receives an optional packet, it will be hard for it to relocate in a crowded zone. In the unlikely event that the following switch possesses a slower caching speed, congestion might easily develop and many packets could be lost. We look into the front-end average cache retention rate and the front-end node area overflow to try to solve these problems.

Congestion Occurrence Rate (COR): The congestion occurrence rate is calculated by the present queue length QL along with residual energy RE. Depending on the COR, node S will send the packet to destination D; otherwise, node S will look for another neighbor next to CH or choose to drop the packet (Table 2).

Data is transferred from CM to CH, where CH conducts basic synthesis and removes unnecessary data. These compiled and inaccurate statistics are sent to the BS. A node changes its network settings and sends a HELLO signal to a nearby node to share its path parameters after it is done delivering data. Each node maintains an archive of data pertaining to its neighbors, which records specifics such as the next node's store status, the state of the cache memory assortment, and the separation between the nodes. When a HELLO message is received, each node in the surrounding region modifies its node information table to keep track of the degree of traffic in the area. This drastically reduces instances of network overload as the node chooses a route for every packet's following change.

Algorithm for Congestion Aware Routing using Artificial Bee Colony

Input: Source Node S, Destination Node D

Output: Congestion Free Best Route CFBR
Variables:
Forward Rate-FR
Network Load-NL
Cache Remaining Rate-CRR
Forward Average Cache Remaining Rate-FACRR
Congestion Occur ace Rate-COR
Initialize population of bees.

```
for j= highest quantity of iterations
    for every bees in the population
        Repeat until bees finish the tour to the destination.
        Pick the consecutive food source with probability value P
        Update probability P using P= FR + NL + CRR + FACRR + COR.
    end for
    Update best route based on high P
end for
```

5 Performance Analyses

The trial was simulated, and NS2 was used to assess the proposed ButBeeRoute method. The TER, CCOR, and CARA algorithms are instances of comparison algorithms. The final outcomes are expressed by the results of many experiments. The simulation uses a space of 100 by 100 square meters while it is in use. Then, throughout that area, a few wireless sensor nodes are dispersed at random. The sink is situated in the center of the simulation area, and the node sends the self-gathered data thereafter. We made the assumption that each node generated data at an identical rate in order to simplify the simulation. Table 3 displays the specific simulation parameters. We adjusted a few variables throughout the simulation to meet our various needs.

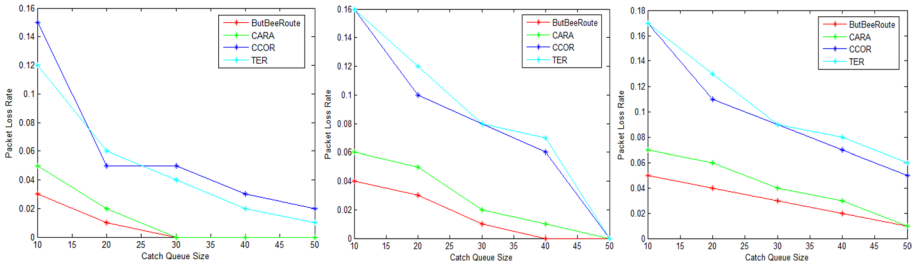
We used the following three route equation metrics in the performance analysis:

(1) The Packet Loss Rate: This indication shows how many packets are lost as a percentage of all the packets sent by the output node.

(2) The Average Number of Hops: The average number of nodes that the output node encounters when transmitting packets to the receiver is represented by this indicator.

Table 3 Simulation parameters & its measurements

Parameters	Measurements
Number of nodes	50–300
area size	100 m x 100 m
Base station coordinate	50,50
Communication range	30 m
Cache queue length	50 packets
Packet size	1024 bits
Buffer size	20 packets
Data rate	4096 bit round
Traffic pattern	Constant Bit Rate
Simulation time	400 rounds



(a) (b) (c)

Fig. 3 **a** Evaluation output of packet delivery ratio for 100 nodes. **b** Evaluation output of packet delivery ratio for 200 nodes. **c** Evaluation output of packet delivery ratio for 300 nodes

(3) The Average Energy Consumption: The average power used per cycle across the network is shown by this indication.

5.1 The Packet Loss Rate

We simulated the rate of network packet loss, which changes based on the size of the node cache, as shown in Fig. 3, to evaluate the ButBeeRoute algorithm's effectiveness in reducing network congestion.

The three sub-graphs in Fig. 3 each have 100, 200, and 300 nodes. Due to the increased number of relay tasks that each network node must complete, packet loss rates rise as the number of nodes grows. The ButBeeRoute technique, however, has a lower packet loss rate than the other two algorithms at various queue lengths, cache sizes, and node counts. This is due to the fact that the CARA, CCOR, and TER algorithms all generate path selections based purely on the node's traffic load. The ButBeeRoute method takes into account both the maximum and average remaining cache speeds.The congested state of the nodes and the anterior region can be accurately reflected by either of these measures. Particularly in terms of the quantity of nodes, our algorithm aims to avoid overweight nodes and clearer areas.

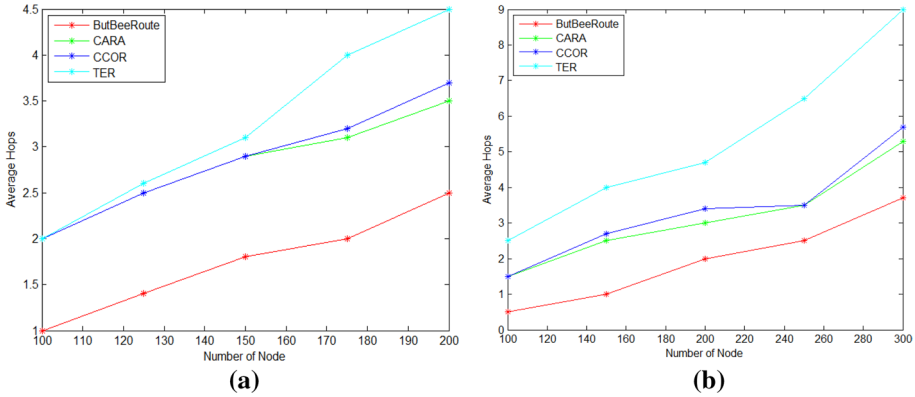


Fig. 4 **a** Evaluation of average number of hops for 200 nodes. **b** Assessment of average number of hops for 300 nodes

5.2 The Average Number of Hops

Network transmission delay is another crucial estimation metric for transmission algorithm estimation. As a result, simulated packets were performed at various node numbers for the caches of nodes 20 and 30. Figure 4 illustrates how the average bounce varies with node density. With the expansion of the network nodes, the three approaches' average number of leaps increases. Both ButBeeRoute and CARA take into account the path length and distance for a single exchange when the node cache size is 20 packs. As a result, the algorithm's average number of hops is small.

Because it aims to make judgments instantly while limiting the amount of distance from the jump, the TER algorithm has the highest average jump rate, which leads to a greater average jump number. Due to this, many packages are dropped, which adds to traffic congestion. Increase the node cache size to 30 packets, and the network loss rate is significantly reduced. This is as a result of one hop's transmission distance factor dictating the nodes to be selected for the next switch.

5.3 The Average Energy Consumption

The suggested ButBeeRoute transmission algorithm does not assess the power consumption while choosing a node for the next switch because the focus of this paper is a wireless sensor network with adequate power. We evaluated and examined the network's overall power consumption because it is a crucial metric for assessing WSNs. The average power consumption is simulated for various nodes when the cache range length is set to 20 and 30 packs, as illustrated in Fig. 5.

With more network nodes, all three methods' average power usage increases. Figure 5 illustrates the TER algorithm's greatest average use of power. It has a high average amount of hops, which leads to maximizing its typical power consumption; therefore, it is unaware of congestion.

The below Table 4 provides a comparison of different methods (Angurala et al. [4], Majeed et al. [22], Amit Grover et al. [23], and Proposed) in a wireless sensor network (WSN) based on three performance metrics: average number of hops, average packet loss

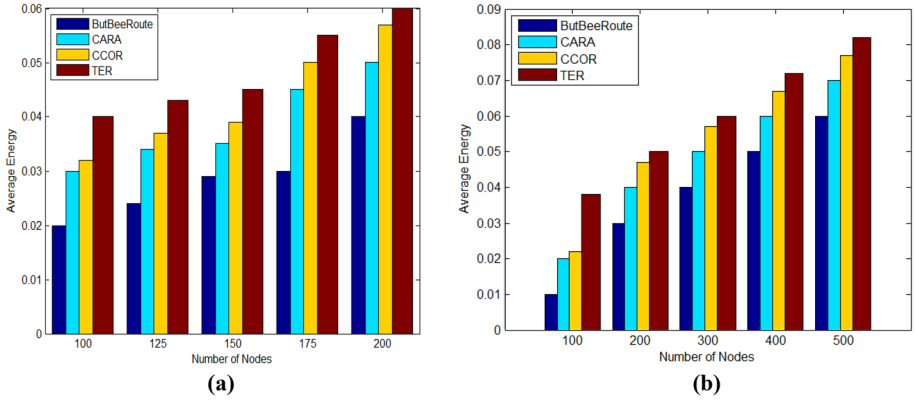


Fig. 5 **a** Assessment of average energy consumption for 200 nodes. **b** Evaluation of average energy consumption for 300 nodes

Table 4 Comparison of different methods

Methods	Average number of hops (%)	Average packet loss ratio (%)	Average energy consumption (%)
Angurala et al. [4]	38	39	42
Majeed et al. [22]	25	30	28
Amit Grover et al. [23]	23	28	24
Proposed	17	20	12

ratio, and average energy consumption. The values for each method represent the percentages of these metrics achieved in their respective evaluations.

From the above table, it is clear that the proposed approach provides a 17% average number of hops, which is lower than the other existing approach. So it is considered to be the proposed approach. Lower values for the average number of hops are desirable as they indicate more efficient routing, reducing the number of intermediate nodes data packets need to traverse.

As well, it is evident that the "proposed method" outperforms the other three methods in all three performance metrics. It achieves the lowest packet loss ratio (20%) and consumes the least amount of energy (12%), making it the most efficient and effective approach for congestion control among the methods evaluated in the WSN.

6 Conclusion

WSN congestion control is a difficult task. Congestion management strategies are more difficult and complex to create due to resource constraints. The ButBeeRoute technique is described in this article. However, based on primary and secondary design goals, BeeRoute approaches are classified as central and dispersed strategies. This method is described in depth and assessed using performance metrics and several congestion

measurement designs. The ButBeeRoute algorithm optimizes the transfer of data over the network by detecting the feeling of overcrowding in the sensor system and nearby locations. Experimental results show that the ButBeeRoute method surpasses more well-known methods. However, BeeRoute has the longest life expectancy because it reduces power consumption and packet loss by removing congestion. Finally, any modifications to the transmission model of the actual WSN network are found and examined using the ButBeeRoute algorithm. The proposed work was executed with 300 nodes. In the future, it will be executed with 1000 nodes in the same scenario, and the performance will be evaluated. A supervised learning approach based on congestion management will be considered to enhance performance in the future.

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Data Availability Data will be made available on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

Conflict of interest All the authors declare no conflict of interest to report regarding the present study.

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